

Logistic Regression: The Sigmoid Function

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Classification problems ([Note 3](#)); Model-based learning ([Note 6](#)); Gradient descent ([Note 57](#)); Polynomial features ([Note 61](#)); Perceptron trick ([Note 71](#)).
- **Leads to:** Softmax regression ([Note 79](#)); Gradient boosting ([Note 120](#)); Multi-layer perceptron (MLP) ([Note 1003](#)); Backpropagation ([Note 1015](#)); Vanishing gradient ([Note 1018](#)).
- **Compare with:** Support vector machines ([Note 92](#)).

1. Overview

Key point: The perceptron trick stops as soon as every point is correct. The fix: let every point act on the line, with strength depending on its distance. Replacing the step function with the sigmoid function does exactly this, and also turns the model's output into a probability.

The previous Note found the weakness of the perceptron trick: once no point is misclassified, the line stops moving, wherever it happens to be. Logistic regression keeps improving the line after that.

This Note changes the perceptron's rule so that correctly classified points also take part. The change needs one new ingredient, the **sigmoid function**, one of the most important functions in machine learning and deep learning.

2. The new idea: every point pushes or pulls

Key point: Misclassified points pull the line towards themselves; correctly classified points push it away. The strength depends on how far the point is from the line.

2.1 Push and pull

Key point: Old rule: only misclassified points act. New rule: all points act.

So far, only misclassified points acted: each one **pulled** the line towards itself. Correct points did nothing.

The new rule: correctly classified points **push** the line away from themselves. Then, even when every point is correct, the line keeps moving: points on both sides push it, and it keeps moving until the pushes balance. The aim is a line in the middle of the gap; Section 7 measures how close this rule gets.

2.2 How strongly

Key point: Near correct points push hard, far ones gently. Near misclassified points pull gently, far ones hard.

The strength of each push or pull depends on the point's distance from the line:

Point	Near the line	Far from the line
Correctly classified	strong push	weak push
Misclassified	weak pull	strong pull

This makes sense. A correct point right next to the line is at risk, so it should push hard. A correct point far away is safe, so it hardly needs to act. A badly misclassified point, far on the wrong side, is a big error, so it should pull hard.

3. Why the step function cannot do this

Key point: With the step function, $\hat{y}$ is 0 or 1, so for every correct point $y - \hat{y} = 0$ and the update vanishes.

The update rule from the perceptron Notes is

$$w_{\text{new}} = w_{\text{old}} + \eta (y - \hat{y}) x$$

For a correctly classified point, y and $\hat{y}$ are equal (both 0 or both 1), so $y - \hat{y} = 0$ and nothing changes. To let correct points act, $y - \hat{y}$ must not be 0.

y is the true class from the data, so it cannot change. What can change is how the model produces $\hat{y}$. So far it computed $z = w \cdot x$ and passed it through a **step function**: 1 if $z > 0$, otherwise 0. As long as $\hat{y}$ is only ever 0 or 1, the problem remains. We need a function whose output can be anything between 0 and 1.

4. The sigmoid function

Key point: $\sigma(z) = 1 / (1 + e^{-z})$. It squeezes any number into the range 0 to 1, with $\sigma(0) = 0.5$.

The **sigmoid function** is

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

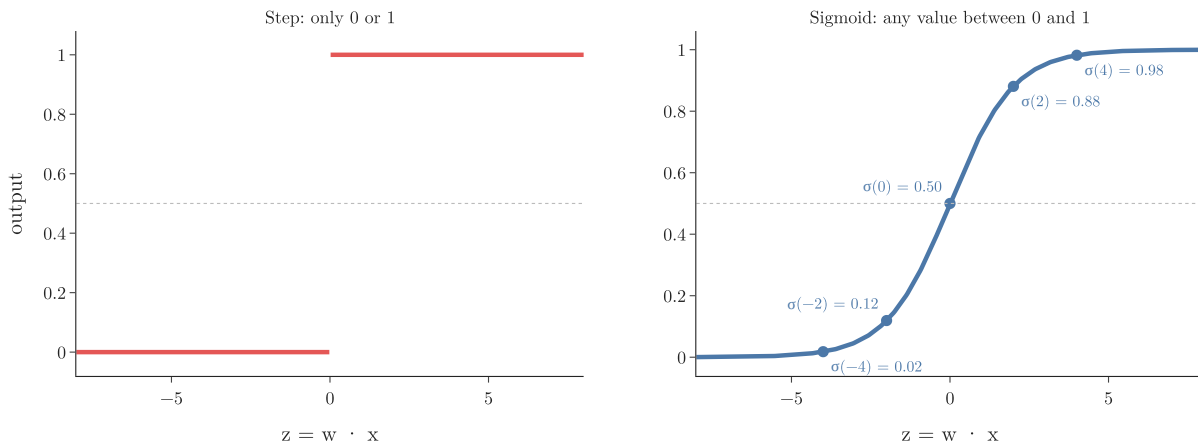


Figure 1: The step function and the sigmoid function

Its key properties (Figure 1, right):

- For a very large z , e^{-z} is almost 0, so $\sigma(z)$ is almost 1.
- For a very negative z , e^{-z} is huge, so $\sigma(z)$ is almost 0.
- At $z = 0$, $\sigma(0) = 1/(1 + 1) = 0.5$.

z	-4	-2	0	2	4
$\sigma(z)$	0.018	0.119	0.5	0.881	0.982

However large or small the input, the output always stays strictly between 0 and 1. It never actually reaches 0 or 1. The S shape gives the function its name: “sigmoid” means “shaped like the Greek letter sigma”, here like an S (Online Etymology Dictionary).

5. Sigmoid as a probability

Key point: $\sigma(w \cdot x)$ can be read as the probability that the point belongs to the positive class. It is 0.5 on the line and rises towards 1 further onto the positive side.

5.1 The new prediction

Key point: Compute $z = w \cdot x$ as before, but pass it to the sigmoid instead of the step.

For a student with CGPA 7.5 and IQ 110, the model still computes $z = w_0 + w_1 \times 7.5 + w_2 \times 110$. But now $\hat{y} = \sigma(z)$:

- $z > 0$ (positive side): $\hat{y} > 0.5$;
- $z < 0$ (negative side): $\hat{y} < 0.5$;

- $z = 0$ (on the line): $\hat{y} = 0.5$.

To make a yes-or-no prediction, we predict “placed” when $\hat{y} \geq 0.5$, which is the same decision as before.

5.2 A map of probabilities

Key point: Lines parallel to the boundary have equal probability; the probability changes gradually across the plane.

Figure 2 colours the plane by $\sigma(x_1 + x_2)$, for the line $x_1 + x_2 = 0$.

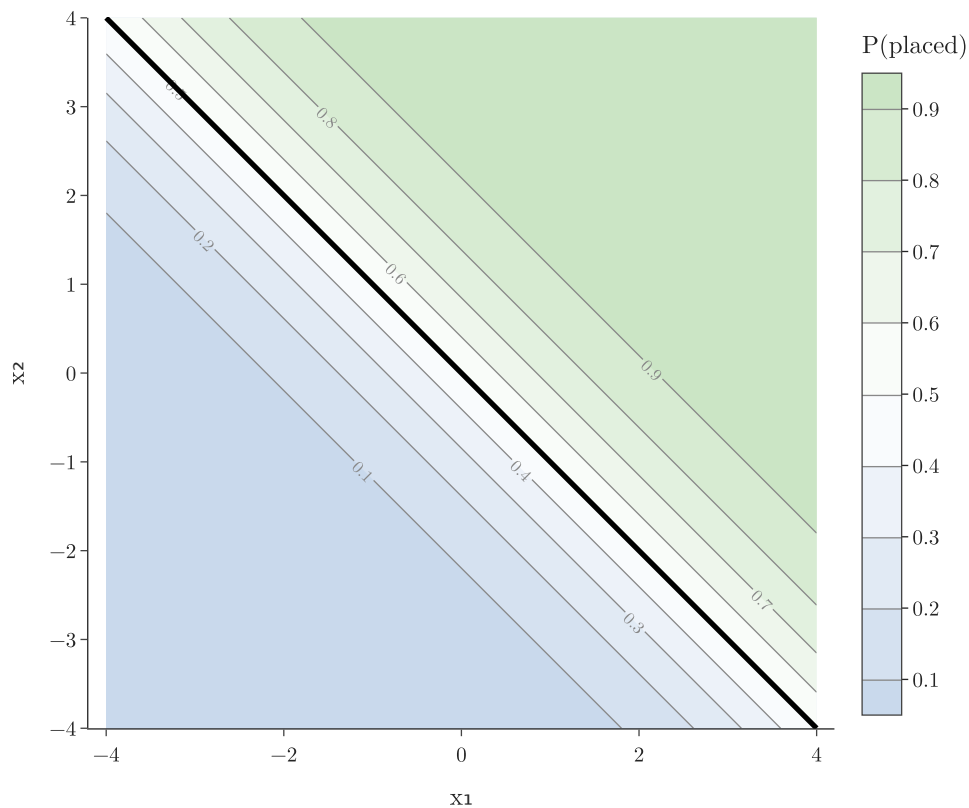


Figure 2: $\sigma(w \cdot x)$ as the probability of being placed

- On the line itself, $\sigma = 0.5$: a 50% chance of being placed.
- Moving onto the positive side, the value rises through 0.6, 0.7, 0.8 and 0.9, along lines parallel to the boundary. Far away it approaches 1.
- On the negative side it falls towards 0.

So instead of a hard yes or no, every student now gets a **probability of being placed**: the further onto the positive side, the higher the chance. The probability of **not** being placed is $1 - \hat{y}$. A student with $\hat{y} = 0.7$ has a 70% chance of being placed and a 30% chance of not.

This probabilistic reading is the second view of logistic regression mentioned in the perceptron Note, and the next Note builds on it.

6. The new update in action

Key point: With $\hat{y} = \sigma(z)$, $y - \hat{y}$ is never exactly 0, so every point moves the line, by an amount that depends on its distance.

6.1 Four points

Key point: Correct points give small values of $y - \hat{y}$ that push; misclassified points give large values that pull.

Take four points and suppose the model gives these probabilities:

Point	True y	Side of the line	$\hat{y} = \sigma(z)$	$y - \hat{y}$	Effect
A	1	positive (correct)	0.80	+0.20	small push
B	0	negative (correct)	0.30	-0.30	push
C	1	negative (wrong)	0.30	+0.70	strong pull
D	0	positive (wrong)	0.65	-0.65	strong pull

None of the values is 0, so every point updates the weights.

- **A:** w grows by $0.2\eta x$. As in the perceptron Note, adding x moves a positive point further onto the positive side: the line moves away from it, a push.
- **C:** w grows by $0.7\eta x$: the same direction but much larger, enough to bring the line towards and past C, a pull.
- **B** and **D** work the same way with subtraction.

6.2 Strength and distance

Key point: $y - \sigma(z)$ is large for points deep on the wrong side and small for points deep on the correct side.

Figure 3 draws $y - \sigma(z)$ against z , the point's signed position relative to the line.

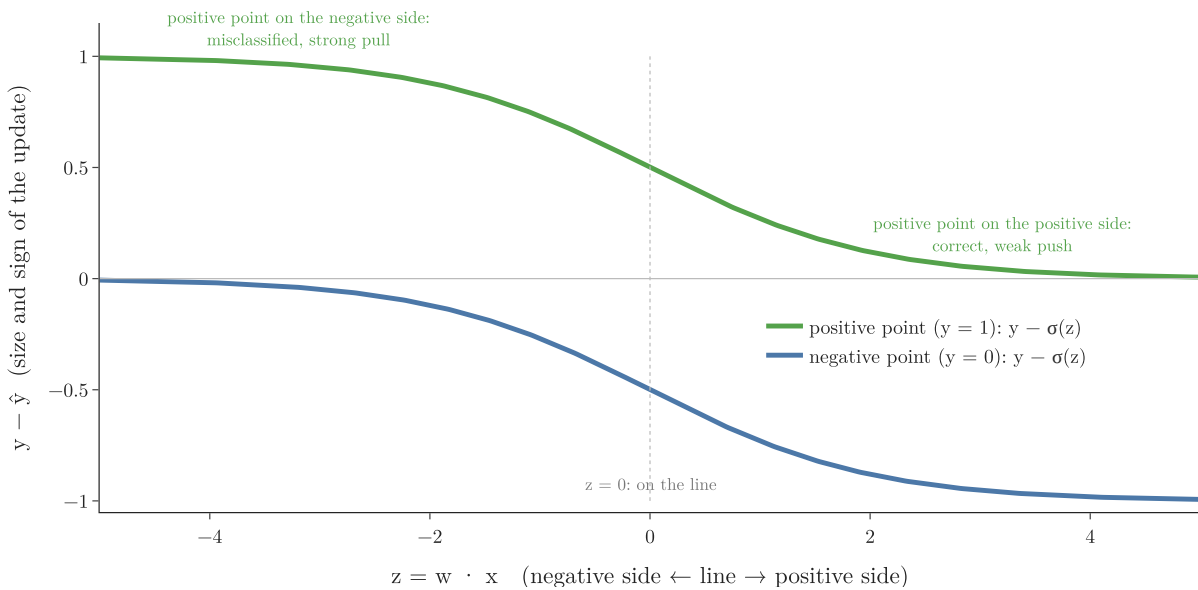


Figure 3: The size of the update against the point's position

For a positive point (green, $y = 1$):

- deep on the wrong side ($z = -4$): $1 - 0.018 = 0.98$, a strong pull;
- on the line ($z = 0$): 0.5 ;
- deep on the correct side ($z = 4$): $1 - 0.982 = 0.018$, a very weak push.

For a correct point, the push is strongest when it is just on the correct side, near the line, and fades as it gets further away. The negative points (blue) mirror this. That is exactly the behaviour planned in the table of Section 2.2.

7. Does it help?

Key point: The sigmoid version places the line better than the step version, but still not as evenly as logistic regression.

Python: The perceptron with a sigmoid.

```
def sigmoid(z):
    return 1 / (1 + np.exp(-z))

def perceptron(X, y, lr=0.1, loops=1000, seed=0):
    rng = np.random.default_rng(seed)
    X = np.insert(X, 0, 1, axis=1)
    w = np.ones(X.shape[1])
    for _ in range(loops):
        j = rng.integers(0, X.shape[0])
        y_hat = sigmoid(X[j] @ w)  # was: step(X[j] @ w)
```

```

w = w + lr * (y[j] - y_hat) * X[j]
return w

```

Figure 4 compares three lines on data with a wide gap between the classes (`make_classification` with `class_sep=30`):

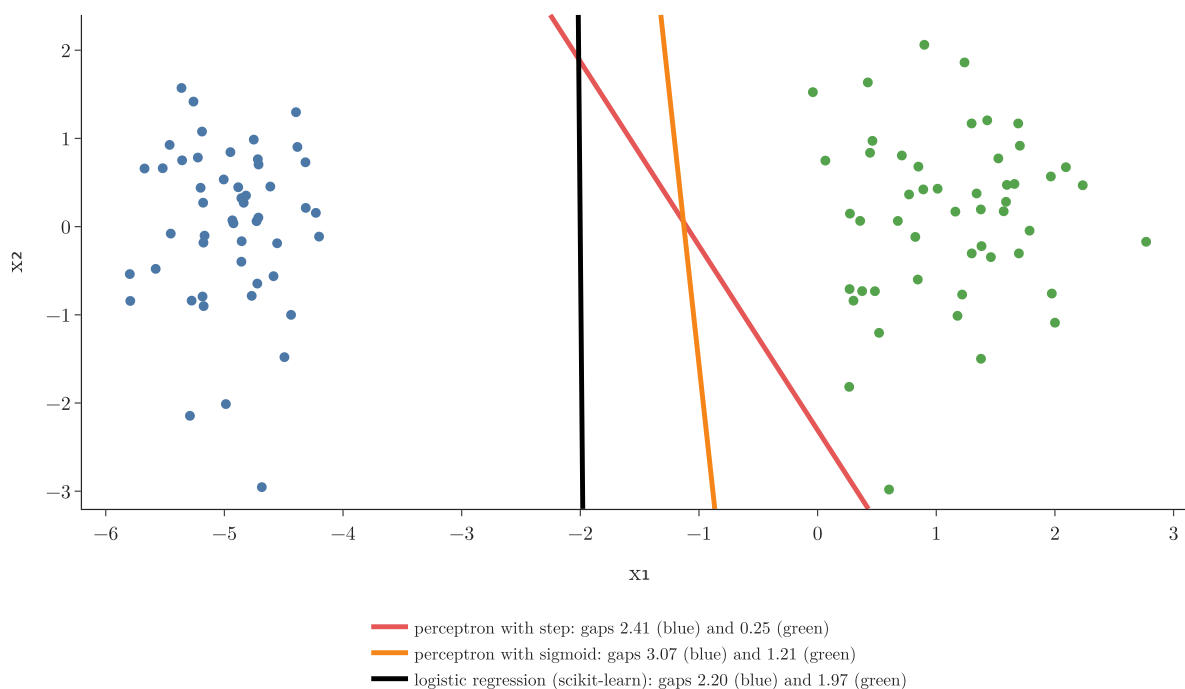


Figure 4: Step perceptron, sigmoid perceptron and logistic regression

Model	Gap to the blue class	Gap to the green class
Perceptron with step	2.41	0.25
Perceptron with sigmoid	3.07	1.21
Logistic regression (scikit-learn)	2.20	1.97

The step version stops right next to the green class. The sigmoid version moves well away from it, a clear improvement. But it is still not centred, while scikit-learn’s logistic regression keeps a nearly equal gap on both sides.

So the change was in the right direction. Why is the sigmoid line still off-centre? The line is not finished yet. Run longer, the sigmoid perceptron keeps moving towards the logistic regression line, and with the small penalty that scikit-learn adds by default it lands on that line (details in the Extra below). In fact the sigmoid update is already the gradient descent step of logistic regression (Bishop §4.3.2), as the [gradient descent Note](#) will show.

What we still lack is a way to say which line is best: a **loss function**, one number that measures how good a line is. With a loss function, we know what the updates are minimising and when to stop. The loss function is the subject of the next Note.

Extra: The test, in the notebook. Gaps (blue, green) of the sigmoid perceptron:

Loops	No penalty	With scikit-learn's default penalty
1,000	3.07, 1.21	2.64, 1.57
100,000	2.61, 1.62	2.19, 1.93
1,000,000	2.48, 1.76	2.18, 2.00

scikit-learn's line has gaps 2.20 and 1.97. Without a penalty, the line keeps drifting slowly: on separable data such as this, plain log loss has no finite best line, and the weights grow for ever (Bishop §4.3.2). The penalty fixes that and gives one definite answer.

8. Summary

- Fix for the perceptron: every point acts; correct points push, misclassified points pull, with strength depending on distance.
- With the step function, $y - \hat{y} = 0$ for correct points, so they cannot act.
- The sigmoid $\sigma(z) = 1/(1 + e^{-z})$ maps any z into $(0, 1)$, with $\sigma(0) = 0.5$.
- $\sigma(w \cdot x)$ is the probability of the positive class: 0.5 on the line, near 1 deep on the positive side.
- Using $\hat{y} = \sigma(z)$ in the update lets every point act and improves the line, but not yet to logistic regression's quality.

9. Sources

- **Online Etymology Dictionary:** “sigmoid”, etymonline.com.
- **Bishop:** Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006. Section 4.3.2, pp. 205–207.

10. Key terms

Term	Meaning
Sigmoid function	$\sigma(z) = 1/(1 + e^{-z})$; an S-shaped curve that maps any number into the range 0 to 1
Logistic function	Another name for the sigmoid function
Probabilistic interpretation	Reading the model's output as the probability of the positive class
Push and pull	Moving the line away from a correctly classified point, or towards a misclassified one